

Module 04 Lab - Exploratory Data Analysis

Objective: To learn how to explore a dataset to find patterns, anomalies, and insights before modeling. EDA is like being a detective; you are looking for clues in the data that will help you build a better model.

This lab is fully coded. Your task is to run each cell, read the detailed explanations, understand the purpose of each visualization, and then complete the experimentation section.

Part 1: Setup and Data Loading

What is Exploratory Data Analysis (EDA)? EDA is the process of using summary statistics and visualizations to understand a dataset's main characteristics. Before you can build a model, you need to understand your data. What stories does it tell? Are there errors or missing values? Are there strong relationships between variables? EDA helps answer these questions. We will use the famous Titanic dataset for this lab. It contains information about passengers and, crucially, whether they survived the disaster.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns # Seaborn is a library built on top of Matplotlib that makes creating beautiful plots easier.
# Load the dataset directly from a URL
df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')
print("---- First 5 Rows ----")
print(df.head())
print("---- Basic Info ----")# .info() is a great first command. It tells us the column names, how many non-null values are in each column, and their data types.# Notice that 'Age' and 'Cabin' have missing values!
df.info()

---- First 5 Rows ----
   PassengerId  Survived  Pclass  \
0             1         0       3
1             2         1       1
2             3         1       3
3             4         1       1
4             5         0       3

   Name                               Sex  Age  SibSp  \
0  Braund, Mr. Owen Harris             male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                               Heikkinen, Miss. Laina  female  26.0      0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                               Allen, Mr. William Henry  male  35.0      0

   Parch  Ticket   Fare  Cabin  Embarked
0      0   A/5 21171  7.2500   NaN        S
1      0  PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282  7.9250   NaN        S
3      0   113803  53.1000  C123        S
4      0   373450  8.0500   NaN        S

---- Basic Info ----
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column              Non-Null Count  Dtype
---  -
 0   PassengerId         891 non-null    int64
 1   Survived            891 non-null    int64
 2   Pclass              891 non-null    int64
 3   Name                891 non-null    object
 4   Sex                 891 non-null    object
 5   Age                 714 non-null    float64
 6   SibSp              891 non-null    int64
 7   Parch              891 non-null    int64
 8   Ticket              891 non-null    object
 9   Fare                891 non-null    float64
10   Cabin              204 non-null    object
11  Embarked            889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

Part 2: Descriptive Statistics

Let's start by getting a high-level numerical summary of the data.

The `.describe()` method is perfect for this. It calculates statistics like mean, standard deviation, min, and max for the numerical columns.

```
# Get summary statistics for numerical columns
print("---- Descriptive Statistics ----")
print(df.describe())
print("---- Key Insights from Statistics ----")
print(f"The average age of a passenger was {(df['Age'].mean():.1f)} years.")
print(f"The overall survival rate was {(df['Survived'].mean():.1%).}")
print(f"Fares ranged from ${df['Fare'].min()} to a whopping ${df['Fare'].max()}.")

---- Descriptive Statistics ----
   PassengerId  Survived  Pclass    Age  SibSp  \
count  891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642  29.699118    0.523008
std    257.353842    0.486592    0.836071  14.526497    1.102743
min      1.000000    0.000000    1.000000    0.420000    0.000000
25%    223.500000    0.000000    2.000000   20.125000    0.000000
50%    446.000000    0.000000    3.000000   28.000000    0.000000
75%    668.500000    1.000000    3.000000   38.000000    1.000000
max    891.000000    1.000000    3.000000   80.000000    8.000000

   Parch    Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std     0.806057   49.693429
min     0.000000    0.000000
25%     0.000000    7.910400
50%     0.000000   14.454200
75%     0.000000   31.000000
max     6.000000  512.329200

---- Key Insights from Statistics ----
The average age of a passenger was 29.7 years.
The overall survival rate was 38.4%.
Fares ranged from $0.0 to a whopping $512.3292.
```

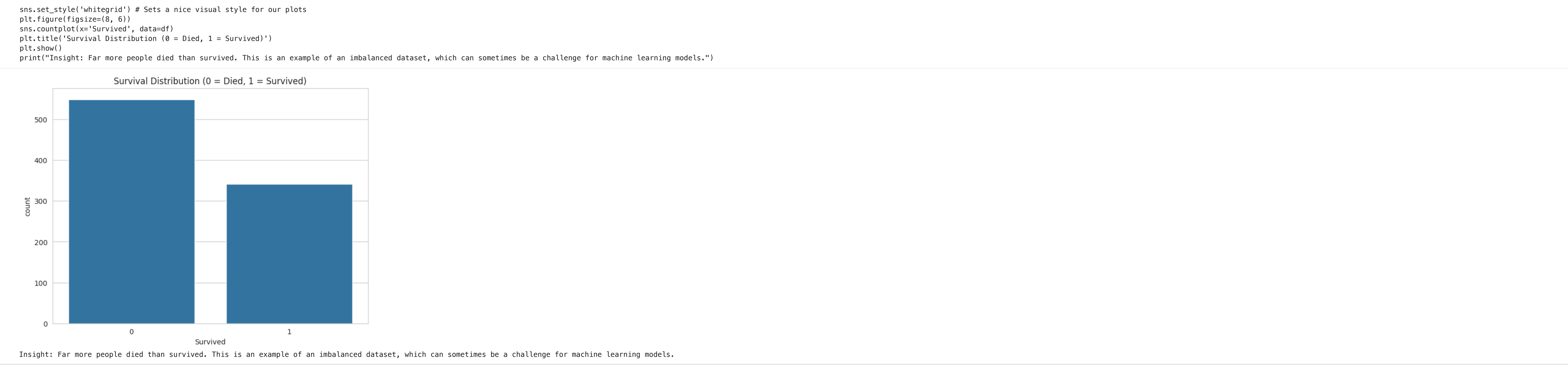
Part 3: Visual EDA - Telling Stories with Plots

Numbers are great, but plots make patterns and relationships immediately obvious. The goal of visual EDA is to turn data into insights.

A Note on Plotting Libraries: **Matplotlib:** The foundational library, gives you full control over everything. **Seaborn:** Built on Matplotlib, it provides a simpler, high-level interface for creating common statistical plots. We will use Seaborn for its ease of use and attractive defaults.

Visualization 1: How many survived?

A simple `countplot` is the best way to see the distribution of a categorical variable, like our target `Survived`.



Visualization 2: Does passenger class matter for survival?

Now we want to see if there's a relationship between two variables. We can use the `hue` parameter in `countplot` to split the bars by another category.



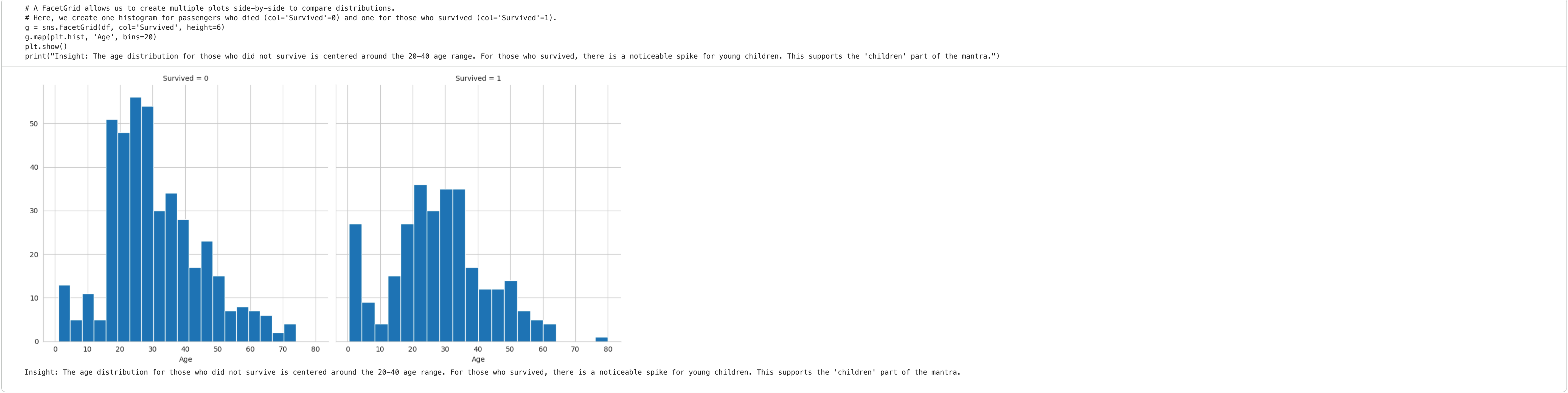
Visualization 3: What about gender?

The 'women and children first' mantra is famous. Let's see if the data supports it.





Visualization 4: How does age play a role?For a continuous variable like `Age`, a histogram is a great way to see its distribution.



Part 4: Student ExperimentationInstructions: Create your own visualizations to explore other relationships in the data.

Experiment 1: Port of Embarkation1. The `Embarked` column tells you where a passenger boarded the ship (C = Cherbourg, Q = Queenstown, S = Southampton).2. Create a `countplot` to see how survival rates differed by the port of embarkation. Does where they boarded seem to be related to their survival?



Experiment 2: Fare vs. Survival1. `Fare` is a continuous numerical variable.2. A `boxplot` or `violinplot` is a great way to see the distribution of fares for those who survived vs. those who didn't.3. Create one of these plots to compare the `Fare` distribution by `Survived`. What does it tell you?



🧠 Knowledge CheckInstructions: Answer the following questions in this markdown cell.1. What is the primary goal of Exploratory Data Analysis (EDA)?2. Based on the plots in this lab, what kind of person had the best chance of surviving the Titanic? (Describe them in terms of class, gender, and age).3. Why is it important to visualize data instead of just looking at summary statistics? What can a plot show you that a number like 'mean' or 'count' can't?[ENTER YOUR ANSWERS HERE]

- What is the primary goal of Exploratory Data Analysis (EDA)?

The main goal of EDA is to explore and understand the data before doing more analysis or building a model. It helps us find patterns, relationships, and anything unusual in the data.
- Based on the plots in this lab, what kind of person had the best chance of surviving the Titanic?

Based on the plots, women in first class had the best chance of surviving. Younger passengers, especially children, also had a better chance of survival.
- Why is it important to visualize data instead of just looking at summary statistics?

Visualizing data makes it easier to see patterns and differences between groups. A mean or count only gives us a number, but a plot can show the distribution, outliers, trends, and relationships in the data that we might not notice from numbers alone.

